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EXAMINER	
DIVECHA, KAMAL B	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/570,018	Applicant(s) Fang et al.	
	Examiner KAMAL B DIVECHA	Art Unit 2453	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03/04/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20,23-29,31-32 and 34-37 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20,23-29,31-32 and 34-37 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 03/04/26.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This action is in response to communications filed 03/04/2026.

Claims 1-20, 23-29, 31-32, 34-37 are pending and presented for examination.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 3/4/2026 is being considered by the examiner.

Continued Examination Under 37 CFR 1.114

A request for continued examination under 37 CFR 1.114, including the fee set forth in 37 CFR 1.17(e), was filed in this application after final rejection. Since this application is eligible for continued examination under 37 CFR 1.114, and the fee set forth in 37 CFR 1.17(e) has been timely paid, the finality of the previous Office action has been withdrawn pursuant to 37 CFR 1.114. **Applicant's submission filed on 03/04/26 has been entered.**

Response to Arguments

Applicant's arguments with respect to claim(s) have been considered but are moot because the new ground of rejection does not rely on any reference applied in the prior rejection of record for any teaching or matter specifically challenged in the argument.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claim(s) **1-20, 23-26, 28-29, 31-32, 34, 36-37** is/are rejected under **35 U.S.C. 103** as being unpatentable over **Han et al. (hereinafter Han, US 2018/0076970 A1**) in view of **Shearer et al. (US 2006/0018249 A1**).

As per claim 1, Han discloses a system comprising:

a vehicle having a first network of a first type and a second network of a second type, wherein the first type is different from the second type [**fig. 2: Ethernet network connected to gateway which is connected to CAN network, [0050], [0063]];**

a first device on the first network **[fig. 2: [0063]];** and

a converged network device (CND) interposed between the first network and the second network [**fig. 2: Ethernet network connected to gateway which is connected to CAN network, [0050], [0063]];** and structured to:

facilitate communications between the first device and the second network **[fig. 2, 9, 10, [0063], [0100-0112]].**

However, Han does not teach

up-sample one or more messages between the first network of the first type and the second network of the second type,

wherein:

the up-sample inserts additional messages into the one or more messages to generate up-sampled data messages, and

the additional messages include interpolations of original values of the one or more messages.

Shearer, from the same field of endeavor, teaches

up-sample one or more messages between the first network of the first type and the second network of the second type [**claim 12, claim 22],**

wherein:

the up-sample inserts additional messages into the one or more messages to generate up-sampled data messages **[claim 12, claim 22**], and

the additional messages include interpolations of original values of the one or more messages **[claim 12, claim 22, claim 1 and claim 10], [0058], [0035], [0007]].**

Therefore, it would have been obvious to a person of ordinary skilled in the art before the effective filing date of the claimed invention to modify Han in view of Shearer in order to up-sample one or more messages between the first network and second network by interpolations of original values of the one or more original messages.

One of ordinary skilled in the art would have been motivated because increasing the data rate and allowing more effective use of bandwidth for devices operating in different bands enables more efficient communications and also higher data rate enables service providers to more effectively use their allotted spectrum [**Shearer: [0007-0008]].**

As per claim 2, Han-Shearer discloses the system of claim 1, further comprising: a second device on the second network and structured to communicate with the first device via the CND [**Han: fig. 2, 9, 10, [0063-0064], [0100-0112]].**

As per claim 3, Han-Shearer discloses the system of claim 1 further comprising: an external device communicatively coupled to the CND, wherein the external device is structured

to adjust a configuration of the CND **[Han: [0088-0091]: 3 modes of operation for the hardware layer. When in sleep mode, it can be waken up when a remote event is detected, i.e. external event from external client or device is received, which changes the configuration mode, inactive, sleep or wake-up of the device**].

As per claim 4, Han-Shearer discloses the system of claim 1, wherein at least one of the first network and/or the second network is a Controller Area Network (CAN) based network [**Han: fig. 2, 9, 10, [0063], [0100-0112]].**

As per claim 5 , Han-Shearer discloses the system of claim 1, wherein at least one of the first network **and/or** the second network is an Ethernet based network **[Han: [fig. 2, 9, 10, [0063-0064], [0100-0112]].**

As per claim 6, Han-Shearer discloses the system of claim 5, wherein the Ethernet based network comprises a data bus architecture **[Han: fig. 1-fig. 2, [0063-0064], [0078]: Bus topology].**

As per claim 7, Han-Shearer discloses the system of claim 1, wherein the first network is a Controller Area Network (CAN) based network and the second network is an Ethernet based network [**Han: fig. 2, 9, 10, [0063-0064], [0100-0112]].**

As per claim 8, Han-Shearer discloses the system of claim 7, wherein the CND comprises:

a communication device structured to access the CAN based network **[Han: Fig. 2 gateway 200, [0063]];**

an Ethernet switch structured to access to the Ethernet based network **[Han: fig. 2 gateway 200, [0093-0098]];** and

wherein the communication device provides a message to the Ethernet switch in response to a corresponding message on the CAN based network **[[0093-0098], fig. 10, fig. 16, [0128-0131]].**

As per claim 9, Han-Shearer discloses the system of claim 8, wherein the communication device is structured to provide the message to the Ethernet switch as an Ethernet message including at least one encapsulated CAN message based on the corresponding message on the CAN based network [**Han:** [**0093-0098], fig. 10, fig. 16, [0128-0131]].**

As per claim 10, Han-Shearer discloses the system of claim 9, wherein the at least one encapsulated CAN message includes more than one encapsulated CAN message [**Han: fig. 3, fig. 14, fig. 15, [0106-0112], [0118]: multiple DU's are included].**

As per claim 11, Han-Shearer discloses the system of claim 8, wherein the communication device is positioned in a first housing, and wherein the Ethernet switch is positioned in a second housing **[Han: fig. 9: CAN PHY layer and Ethernet PHY layer units are separate units/chips, [0080]].**

As per claim 12, Han-Shearer discloses the system of claim 8, wherein the communication device and the Ethernet switch are positioned in a single housing [**Han: fig. 2: Gateway 200, fig. 9, [0080]].**

As per claim 13, Han-Shearer discloses the system of claim 7, wherein the CND comprises: a communication device structured to access the CAN based network; an Ethernet switch structured to access to the Ethernet based network; and wherein the communication device receives a message from the Ethernet switch in response to a corresponding message on the Ethernet based network [**Han: fig. 2, [0063-0068], fig. 9-10, fig. 16].**

As per claim 14, Han-Shearer discloses the system of claim 13, wherein the communication device is structured to receive the message from the Ethernet switch as an Ethernet message including at least one encapsulated CAN message based on the corresponding message on the Ethernet based network **[Han: [0093-0098], fig. 10, fig. 16, [0128-0131]].**

As per claim 15, Han-Shearer discloses the system of claim 14, wherein the at least one encapsulated CAN message includes more than one encapsulated CAN message **[Han: fig. 3, fig. 14, fig. 15, [0106-0112], [0118]: multiple DU's are included].**

As per claim 16, Han-Shearer discloses the system of claim 13, wherein the communication device is positioned in a first housing, and wherein the Ethernet switch is positioned in a second housing **[Han: fig. 9: CAN PHY layer and Ethernet PHY layer units are separate units/chips, [0080]].**

As per claim 17, Han-Shearer discloses the system of claim 13, wherein the communication device and the Ethernet switch are positioned in a single housing **[Han: fig. 2: Gateway 200, fig. 9, [0080]].**

As per claim 18, Han discloses An apparatus comprising:

a first network interface circuit structured to interpret a first network data set having a first network format [**fig. 9-10, fig. 16**];

a translation circuit structured to:

determine one or more message values from the first network data set **[[0106]: payload of the message**] and to encode the one or more message values in a second network data set having a second network format [**fig. 10,**

[0064-0067] CAN message and Ethernet message, [0094]: conversion, [0109-0111]: generate CAN message from the Ethernet message, fig. 16 and [0127-0131]: generate ethernet message from CAN message];

second network interface circuit structured to transmit the second network data set [**fig. 2, fig. 9-fig. 10, fig. 16, 0109-0111], 0127-0131];**

wherein:

the first network format and the second network format are vehicle data formats [**fig. 2: Ethernet network connected to gateway which is connected to CAN network, [0050], [0063], [0094]];** and

the first network format is different than the second network format **[fig. 2: Ethernet network connected to gateway which is connected to CAN network, [0050], [0063], [0094]].**

However, Han does not teach

wherein the one or more encoded message values are up-sampled from the one or more message values via an insertion of additional message values into the one or more message values, and the up-sample adjusts a ratio of a number of the one or more encoded messages values to the one or more message values;

the additional message values include interpolations of original values of the one or more message values.

Shearer, from the same field of endeavor, teaches

wherein the one or more encoded message values are up-sampled from the one or more message values via an insertion of additional message values into the one or more message values, and the up-sample adjusts a ratio of a number of the one or more encoded messages values to the one or more message values [**claim 12, claim 22, [0058]],**

the additional message values include interpolations of original values of the one or more message values [**claim 12, claim 22, claim 1 and claim 10], [0058], [0035], [0007**]].

Therefore, it would have been obvious to a person of ordinary skilled in the art before the effective filing date of the claimed invention to modify Han in view of Shearer in order to up-sample one or more messages between the first network and second network by interpolations of original values of the one or more original messages.

One of ordinary skilled in the art would have been motivated because increasing the data rate and allowing more effective use of bandwidth for devices operating in different bands enables more efficient communications and also higher data rate enables service providers to more effectively use their allotted spectrum [**Shearer: [0007-0008]].**

As per claim 19, Han discloses the apparatus of claim 18, further comprising: a configuration circuit structured to configure the first network interface circuit, the translation circuit, and/or the second network interface circuit, responsive to a configuration command value from a device external to a vehicle, the vehicle including the first network interface circuit and the second network interface circuit [**fig. 2, 10, 16, [0088-0091]: 3 modes of operation for the hardware layer. When in sleep mode, it can be waken up when a remote event is**

detected, i.e. external event from external client or device is received, which changes the configuration mode, inactive, sleep or wake-up of the device, [0067, 0104-0105]: Ethernet message from endpoint in ethernet network includes header, at least one DU. Ethernet message further includes authentication information ASIL such as authentication key, a hash value, CRC, FRC and the like. The header includes destination address DA, source address SA, type filed, IP header, id, flag field, etc. [0129]: CAN message also includes control or management information].

As per claim 20, Han discloses the apparatus of claim 19, wherein the configuration command value comprises **at least one value** selected **from the values consisting of**: a sampling rate value; a payload processing value; a timestamp value; **a metadata value**; **a target device** (i.e. destination) for the one or more message values from the first network data set, **wherein the target device comprises a device on the vehicle** to **send and/or receive** the one or more message values; and **an encapsulation description for the one or more message values [[0067, 0104-0105**]: Ethernet message from endpoint in ethernet network includes header, at least one DU. Ethernet message further includes **authentication information ASIL** such as authentication key, a hash value, CRC, FRC and the like. The header includes **destination address DA**, source address SA, **type filed,** IP header, id, flag field, etc. [0129]: CAN message also includes control or management information].

As per claim 23, Han discloses the apparatus of claim 18, wherein one of the first network format or the second network format is Controller Area Network (CAN) based, and the other of the first network format or the second network format is Ethernet based **[Han: fig. 2, 9, 10, [0063], [0094], [0100-0112]].**

As per claim 24, Han discloses the apparatus of claim 23, wherein one or more CAN messages of the CAN based network can be carried in one Ethernet frame of the Ethernet based network [**Han: fig. 2, 9, 10, [0063], [0094], [0100-0112]].**

As per claim 25, Ha discloses the apparatus of claim 18, wherein the first network interface circuit and/or the second network interface circuit is integrated with a configurable edge gateway (CEG) [[**Han: fig. 2, 9, 10, [0063], [0094], [0101-0112]].**

As per claim 26, Ha discloses the apparatus of claim 18, wherein the first network interface circuit and/or the second network interface circuit is integrated with an Ethernet switch [**fig. 2 gateway 200, [0093-0098]].**

As pe claims 28, it does not teach or further define over the limitations in claim 18. Therefore, claim 28 is rejected for the same reasons as set forth in claim 18.

As per claim 29, Han discloses the method of claim 28, further comprising adjusting a configuration at least one of the encoding or the transmitting in response to a configuration change from an external device [**fig. 2, 10, 16, [0088-0091]: 3 modes of operation for the hardware layer. When in sleep mode, it can be waken up when a remote event is detected, i.e. external event from external client or device is received, which changes the configuration mode, inactive, sleep or wake-up of the device**, [**0067, 0104-0105]: Ethernet message from endpoint in ethernet network includes header and at least one DU. Ethernet message further includes authentication information ASIL such as authentication key, a hash value, CRC, FRC and the like. The header includes destination address DA, source address SA, type filed, IP header, id, flag field, etc. [0129]: CAN message also includes control or management information. Based on these parameters from the endpoints, the encoding [such as convert**

or not] or transmitting [drop when authentication fails] can be performed by the gateway device].

As per claim 31, Han discloses the method of claim 28, wherein encoding the message value in the second network data set comprises encapsulating the CAN message into the Ethernet message [**fig. 16, [0127-0131]].**

As per claim 32, Han discloses the method of claim 31, wherein encoding the message value further comprises processing at least one of a payload or a frame portion of the CAN message [f**ig. 16, [0127-0131]].**

As per claim 34, Han discloses the method of claim 28, wherein the at least one CAN message includes at least a first CAN message and a second CAN message [**fig. 16, fig. 13-fig. 15: multiple DU in each CAN message].**

As per claim 36, Han-Shearer discloses the system of claim 1, wherein the up-sampled data messages include the one or more messages and the one or more additional messages [**Shearer: claim 12, claim 22, [0058]], the one or more additional messages includes at least one of an interpolation or an extrapolation of original values of the one or more messages, and the up-sampled data messages both spanning data values of the one or more messages and non-spanning data values of the one or more messages [Shearer: claim 12, claim 22, claim 1 and claim 10], [0058], [0035], [0007]**]. Same rationale as in claim 1 applies.

As per claim 37, Han-Shearer discloses the system of claim 1, wherein the CND is structured to up-sample the one or more messages to match an up-sampling rate [**Shearer: [0007-0008], claim 12, 22 and claim 1]. Same rationale as in claim 1 applies.**

Claim(s) 27 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Han et al. (hereinafter Han, US 2018/0076970 A1**) in view of **Shearer et al. (US 2006/0018249 A1**) and further in view of **Brewton et al. (US 10,181,059 B1**).

As pe claim 27, Han-Shearer discloses the apparatus of claim 18 as set forth above.

However, Han-Shearer does not teach wherein the first network interface circuit, the translation circuit, and/or the second network interface circuit, are compliant with the Motor Industry Software Reliability Association standard and/or the AUTomotive Open System ARchitecture standard.

Brewton teaches wherein the first network interface circuit, the translation circuit, and/or the second network interface circuit, are compliant with the Motor Industry Software Reliability Association standard and/or the AUTomotive Open System ARchitecture standard [col. 7 L55-65, col. 8 L22-28, fig. 2 – AUTOSAR compatible].

Therefore, it would have been obvious to a person of ordinary skilled in the art before the effective filing date of the claimed invention to modify Han-Shearer in view of Brewton in order to ensure the gateway device is compliant with Motor Industry Software Reliability Association standard or AUTomotive Open system Architecture standard.

One of ordinary skilled in the art would have been motivated in order ensure the gateway device is compatible based on the existing standards.

Claim(s) 35 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Han et al. (hereinafter Han, US 2018/0076970 A1**) in view of **Shearer et al. (US 2006/0018249 A1**) and further in view of **O'Mahony et al. (US 2017/0134537 A1).**

As per claim 35, Han-Shearer discloses the system of claim 1 as set forth above.

However, Han-Shearer does not teach wherein the CND is further structured to down-sample one or more other messages between the first network of the first type and the second network of the second type, wherein the down-sample adjusts a number of the one or more other messages.

O'Mahony teaches CND is further structured to down-sample one or more other messages between the first network of the first type and the second network of the second type, wherein the down-sample adjusts a number of the one or more other messages [**[0079, 0109, 0003, 0099-0101]: decrease the data rate based on decrease in signal strength**].

Therefore, it would have been obvious to a person of ordinary skilled in the art before the effective filing date of the claimed invention to modify Han-Shearer in view of O'Mahony in order to down-sample one or more other messages between the first network of the first type and the second network of the second type, wherein the down-sample adjusts a number of the one or more other messages.

One of ordinary skilled in the art would have been motivated in order to keep data rate compatible with the destination network data rates, thus avoiding congestion or bottlenecks.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

- Chae et al., US 2019/0079842 A1 – Failure Diagnosis and Method for In-Vehicle Control Unit
- HEidari et al., US 6,512,739: Down Conversion within an X-DSL Receiver – Discloses Downstream packets may increase in size as a result of Interpolation.
- Tsuchinaga et al., US 2005/0238013 A1: Interpolation packet generated from preceding and subsequent packets by interpolation.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to KAMAL B DIVECHA whose telephone number is **571-272-5863**. The examiner can normally be reached IFP Normal Hours M-F: 8am-4.30pm EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Colleen Fauz can be reached at 5712721667. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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